

Shuang Li

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Summary

I am currently a Postdoctoral Research Associate in the Department of Electrical and Computer Engineering at the University of Victoria (UVic). My research focuses on wireless communications, integrated sensing and communications (ISAC), machine learning (ML), deep reinforcement learning (DRL), and AI-driven healthcare systems. I am also a Sessional Lecturer for *ECE 403/503: Optimization for Machine Learning* at UVic, with experience in curriculum development, graduate-level teaching, and student mentoring.

I have been recognized for my academic and research excellence through the CSC PhD Fellowship (2021-2025) and the NSERC/SFU Invention to Innovation Skills Training Grant (2025-2026). My long-term goal is to advance sensing-assisted intelligent healthcare communications for lifespan monitoring systems.

Employment History

- **Postdoctoral Research Associate**, University of Victoria, CA 2025/09 ~ Present
Department of Electrical and Computer Engineering
Advisor: Prof. Hong-Chuan Yang
- **Sessional Lecturer**, University of Victoria, CA 2025/09 ~ Present
Department of Electrical and Computer Engineering
Course: *ECE 403/503: Optimization for Machine Learning*

Education

- **Ph.D.**, University of Victoria, CA 2020/09 ~ 2025/08
Department of Electrical and Computer Engineering
- **M.Eng.**, Jilin University, China 2017/09 ~ 2020/07
College of Communication Engineering
- **B.Eng.**, Jilin University, China 2013/09 ~ 2017/07
College of Communication Engineering

Research Interests

- Wireless Communications & Wireless Body Area Networks (WBANs)
- Age of Information (AoI) & Green Communications
- Integrated Sensing and Communication (ISAC) & Multimodal Healthcare Sensing
- Multi-Objective Optimization (MOO) & Data-Driven Decision-Making
- Applied Machine Learning & Deep Reinforcement Learning

Miscellaneous Experience

Awards and Achievements

- **i2I Innovation training Grant**, SFU&NSECR 2025/09 ~ 2026/05
- **Research Assistantship**, University of Victoria 2020/09 ~ Present
- **PhD Fellowship**, China Scholarship Council 2021/05 ~ 2025/05
- **First Class Scholarship**, Jilin University 2019/10
- **Outstanding Student Leader**, Jilin University 2019/10
- **Outstanding Graduate Student Scholarship**, Jilin University 2017/12

Teaching Experience

Instructor, University of Victoria Fall 2025 ~ April 2026

- **Course: ECE 403/503: Optimization for Machine Learning**
 - Designed and delivered lectures on various optimization for machine learning.
 - Developed problem sets, lab exercises, and the midterm exam, final exam.
 - Created grading rubrics and coordinated marking with teaching assistants.
 - Held weekly office hours and student consultations and managed the course website and communications.
 - Supervised and mentored graduate-level course projects.

Teaching Assistant, University of Victoria Spring 2022 ~ Fall 2024

- **Course: ECE 403/503: Optimization for Machine Learning**
- **Course: ECE 496A/519A: 5G Wireless and IoT**
- **Course: ECE 310: Digital Signal Processing I**

Research Experience

Research Associate, University of Victoria, YWAILab Fall 2025 ~ Present

Advisor: Prof. Hong-Chuan Yang

- **Project 1: Integrated Sensing and Communications Using 5G RF Signals**
 - Build Digital Twins of indoor environments using real-world 5G RF signals and wireless channel measurements.
 - Develop Transformer-based models for environment reconstruction, localization, and human activity perception without dedicated sensing infrastructure.

- **Project 2: Data-Oriented Performance Metrics for URLLC and Their Applications**
 - Conducted research on ultra-reliable low-latency communications (URLLC) and AI-driven optimization for next-generation healthcare and IoT networks.
 - Investigated data-oriented performance metrics, and intelligent transmission strategies for URLLC.

PhD Researcher, University of Victoria

Fall 2020 ~ Summer 2025

Supervisor: Prof. Hong-Chuan Yang

- **Topic 1: Energy Efficiency Transmission in WBANs: Optimal Design using DRL**
 - Designed energy-efficient transmission protocols for low-power WBAN devices
 - Developed hybrid supervised/DRL solutions for energy consumption minimization
 - Analyzed the effect of slot duration on energy consumption and develop a guideline for practical WBAN design
- **Topic 2: Freshness-Sensitive IoT Networks: Optimal Scheduling Scheme using DRL**
 - Developed intelligent scheduling agents with real-time decision-making capabilities
 - Proposed a parameterized deep Q -network (PDQN) solution for joint scheduling and blocklength selection to minimize AoI
 - Develop a deep transfer learning (DTL) solution to accelerate learning and learn the optimal scheduling policy for AoI minimization
- **Topic 3: Multi-Objective Optimization in IoT Networks: Trade-off with Multi-Objective Reinforcement Learning**
 - Investigated multi-objective reinforcement learning (MORL) to optimize trade-offs relevant to performance, energy, and system reliability in IoT networks
 - Developed preference-guided MORL algorithms to achieve flexible trade-offs among conflicting objectives

Master Researcher, Jilin University

Fall 2017 ~ Summer 2020

Supervisor: Prof. Fengye Hu

- **Topic: Joint Power Allocation in Classified WBANs with Wireless Power Transfer**
 - Designed a novel classified framework for wireless-powered WBANs
 - Proposed a power allocation scheme to maximize system throughput
 - Conducted performance evaluation and optimization through simulation studies

Research Publications

Journal Articles

[1] S. Li, H. Yu, and H. -C. Yang, "Session-Specific Design for Wireless Transmission over WBANs: Navigating Energy Efficiency and Reliability Tradeoff with Multi-Objective Reinforcement Learning," *IEEE Internet Things J.*, vol. 13, no. 10, pp. 20913-20924, May 2026.

- [2] S. Li, H. Hu, H. -C. Yang, K. Xiong, P. Fan and K. B. Letaief, “AoI Minimization for WP-IoT with PDQN-based Hybrid Offline/Online Learning: A Joint Scheduling and Transmission Design Approach,” *IEEE Trans. Cognit. Commun. Networking*, vol. 12, pp. 4547-4560, Jan. 2026.
- [3] S. Li, H. -C. Yang and F. Hu, “Joint Transmission Mode Selection and Scheduling for AoI Minimization in NOMA-Capable WP-IoT Networks: A Deep Transfer Learning Solution,” *IEEE Trans. Commun.*, vol. 73, no. 8, pp. 5805-5816, Aug. 2025.
- [4] S. Li, H. -C. Yang, F. Xu, H. Hu and F. Hu, “Energy-Efficient Relay Transmission for WBAN: Energy Consumption Minimizing Design with Hybrid Supervised/Reinforcement Learning,” *IEEE Internet Things J.*, vol. 11, no. 10, pp. 17770-17779, May 2024.
- [5] S. Li, F. Hu, Z. Xu, Z. Mao, Z. Ling and H. Liu, “Joint Power Allocation in Classified WBANs With Wireless Information and Power Transfer,” *IEEE Internet Things J.*, vol. 8, no. 2, pp. 989-1000, Jan.15, 2021.
- [6] S. Li, F. Hu, Z. Mao, Z. Ling and Y. Zou, “Sum-Throughput Maximization by Power Allocation in WBAN With Relay Cooperation,” *IEEE Access*, vol. 7, pp. 124727-124736, Aug. 2019.

Conference Proceedings

- [1] S. Li, H. Yu and H. -C. Yang, “Towards Energy-Efficient and Freshness-Sensitive WBANs Through Joint Scheduling and Power Control via Multi-Objective Reinforcement Learning”, in *Proc. IEEE WCNC2026*, Presented, Kuala Lumpur, Malaysia, Apr. 2026.
- [2] S. Li, H. Yu and H. -C. Yang, “Multi-Objective Optimization for Energy-Efficient and Reliable Transmission in WBANs: Session-Specific Design Using MODRL,” in *Proc. IEEE VTC2025-Spring*, Oslo, Norway, 2025, pp. 1-5.
- [3] S. Li, F. Xu and H. -C. Yang, “Reliable and Energy-Efficient Relay Transmission in WBANs with Wireless Power Transfer: Optimal Design with DRL,” in *Proc. IEEE PACRIM*, Victoria, BC, Canada, 2024, pp. 1-6.
- [4] S. Li, H. -C. Yang and F. Hu, “Average AoI Minimization in WP-IoT Networks: Optimal Scheduling for NOMA Transmission,” in *Proc. IEEE/CIC ICC*, Dalian, China, 2023, pp. 1-6.
- [5] S. Li, F. Hu, Z. Mao, H. Liu and Z. Ling, “Joint Power Allocation for Energy Harvesting to Maximize Throughput in Classified WBAN,” in *Proc. IEEE GLOBECOM*, Waikoloa, HI, USA, 2019, pp. 1-6.
- [6] S. Li, F. Hu, J. Yu and Z. Huang, “Optimal Power Allocation with a Cooperative Relay in Multi-point WBAN,” in *Proc. IEEE/CIC ICC*, Changchun, China, 2019, pp. 919-924.

Presentations and Seminars

Presentations

- S. Li and H. -C. Yang, “When Will URLLC Benefit from Retransmission? A Study Based on a New Data-Oriented Performance Metric”, São Miguel, Azores, Portugal, May 2026. **(Poster presentation)**
- S. Li, F. Xu and H. -C. Yang, “Reliable and Energy-Efficient Relay Transmission in WBANs with Wireless Power Transfer: Optimal Design with DRL,” Victoria, BC, Canada, Aug. 2024. **(Invited oral presentation on conference paper)**

- S. Li, H. -C. Yang and F. Hu, “*Average AoI Minimization in WP-IoT Networks: Optimal Scheduling for NOMA Transmission*,” Dalian, China, Aug. 2023. (**Invited oral presentation on conference paper**)
- S. Li, F. Hu, Z. Mao, H. Liu and Z. Ling, “*Joint Power Allocation for Energy Harvesting to Maximize Throughput in Classified WBAN*,” Waikoloa, HI, USA, Dec. 2019. (**Poster presentation**)
- S. Li, F. Hu, J. Yu and Z. Huang, “*Optimal Power Allocation with a Cooperative Relay in Multi-point WBAN*,” Changchun, China, Aug. 2019. (**Invited oral presentation on conference paper**)

Seminar

- S. Li, “*Joint Transmission Mode Selection and Scheduling for AoI Minimization in NOMA-Capable IoT Networks*”, University of Victoria, BC, Canada, Apr. 2025. (**UVic ECE Research Day**)
- S. Li, “*Multi-Objective Optimization for Energy-Efficient and Reliable Transmission in WBANs*”, University of Victoria, BC, Canada, Jul. 2025. (**UVic Graduate Student Seminar**)

Professional Service

- **Technical Program Committee Member:**
IEEE GLOBECOM 2025/2026
IEEE PACRIM 2024
IEEE/CIC ICC 2019
- **Reviewer:**
IEEE WCNC 2025
IEEE Communications Magazine
IEEE Internet of Things Journal (IoT)
IEEE Transactions on Vehicular Technology (TVT)
IEEE Transactions on Cognitive Communications and Networking (TCCN)
- **Student Volunteer:**
2024 IEEE Pacific Rim Conference on Communications, Computers and Signal Processing (PACRIM)
2019 IEEE/CIC International Conference on Communications in China (ICCC)

Teaching Interests

- **Wireless & IoT & Signal Processing:** Wireless Communications; Communication Theory; Computer Network; IoT Advanced Engineering Design; Information & Coding Theory; Digital Signal Processing; Signals & Systems
- **Machine Learning & Computing:** Optimization for Machine Learning; Convex Optimization; Deep Reinforcement Learning; Computer Architecture; Python; C++
- **Data Analysis: Data Mining;** Algorithms and Data Models; System for Massive datasets

- **Mathematics & Theory:** Advanced Mathematics; Probability Theory & Statistics; Analytical Methods in Electrical and Computer Engineering; Random Signal Analysis; Introduction to Control Theory

Skills

- **Programming & Frameworks:** Python, C/C++, Java, TensorFlow, PyTorch
- **ML/AI:** Deep learning, reinforcement learning, transfer learning
- **Data Science & Engineering:** Time-series analysis, data analytics, data preprocessing, feature engineering, model evaluation
- **AI Systems:** Real-time AI systems, intelligent decision-making, optimization, deployment-oriented research
- **Software & Tools:** HTML, JavaScript, LaTeX, Overleaf
- **Languages:** English, Mandarin Chinese